

Deep Learning – Practice Paper 1

“Foundation Builder”

24 Questions — 50 Marks — Suggested Time: 2 hours

Covers: Weeks 1–12 — Difficulty: Moderate (matches real exam)

IITM BS Pattern

Type	Count	Marks
MCQ (single correct)	10	20
MSQ (multiple correct)	5	13
SA (numerical)	9	17
Total	24	50

Q1 [MCQ – 2 marks]

Consider a two-dimensional image of size $L \times L$ and a square filter of size $F \times F$. How much padding P is needed on each side to keep the output equal to the input size? Assume stride = 1.

- A. $(F + 1)/2$
- B. $(F - 1)/2$
- C. $F/2$
- D. F

Q2 [MCQ – 2 marks]

Which of the functions has the steepest slope at $x = 1$?

- A. $\log_{10}(x)$
- B. $\ln(x^2)$
- C. $\sqrt{x}$
- D. x^3

Q3 [MCQ – 2 marks]

A neural network has all weights zero, no biases, sigmoid activation in all layers including output. What is the output for any input?

- A. 0
- B. 0.25
- C. 0.5
- D. Depends on input

Q4 [MCQ – 2 marks]

Same network as Q3, but the output activation is changed to **linear**. What is the output?

- A. 0
- B. 0.5
- C. 1
- D. Depends on input

Q5 [MCQ – 2 marks]

Statement 1: A vanilla RNN has more trainable parameters than an LSTM with the same dimensions.

Statement 2: An LSTM has more parameters because it has multiple gates with separate weight matrices.

- A. Statement 1 is true, Statement 2 is the correct reason
- B. Statement 1 is true, Statement 2 is false
- C. Statement 1 is false, Statement 2 is true
- D. Both statements are false

Q6 [MCQ – 2 marks]

Step decay: $\eta_0 = 0.1$, halved every 200 mini-batch updates. Batch size = 50, dataset = 10,000 samples. What is η at the end of epoch 1?

- A. 0.1
- B. 0.05
- C. 0.025
- D. 0.0125

Q7 [MCQ – 2 marks]

In masked self-attention with sequence length T , what is the sum of all diagonal elements of the mask matrix M ?

- A. $-\infty$
- B. ∞
- C. 0
- D. T

Q8 [MCQ – 2 marks]

Which statement about multi-head cross attention is **NOT** true?

- A. Keys and values come from the encoder
- B. Queries come from the decoder
- C. All heads share identical projection matrices
- D. Each head learns different representation subspaces

Q9 [MCQ – 2 marks]

Sigmoid: $\sigma(x) = 1/(1 + e^{-(wx+b)})$, $w > 0$. Increasing b will:

- A. Shift the sigmoid LEFT
- B. Shift the sigmoid RIGHT
- C. Increase the slope
- D. Decrease the slope

Q10 [MCQ – 2 marks]

A XOR function for input (x_1, x_2) uses: one hidden layer with 2 neurons (ReLU), one output neuron (linear). With $W^{(1)} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$, $W^{(2)} = [1 \quad 1]$, and threshold at 0.5, can this network correctly classify XOR?

- A. Yes
- B. No

Q11 [MSQ – 2 marks]

Training error = 8%, test error = 52%. Which methods help reduce test error? (Select all correct.)

- A. Adam optimizer
- B. L2 regularization
- C. Dropout
- D. Increasing number of layers
- E. Data augmentation

Q12 [MSQ – 3 marks]

Which optimizers use momentum or a momentum-like mechanism?

- A. NAG
- B. AdaGrad
- C. RMSProp
- D. Adam
- E. AdaDelta

Q13 [MSQ – 3 marks]

Which problems of the co-occurrence matrix are addressed by SVD? (Select all correct.)

- A. High dimensional representation
- B. Sparse matrix with many zeros
- C. Captures latent semantic similarity between words
- D. Increases vocabulary size

E. Produces dense low-dimensional embeddings

Q14 [MSQ – 3 marks]

Which statements about scaled dot-product attention are correct?

- A. Scaling by $\sqrt{d_k}$ prevents dot products from becoming too large when d_k is high
- B. In multi-head attention, all heads share the same learned projections of Q, K, V
- C. If all heads learn identical projections, MHA behaves like single-head attention
- D. Increasing heads always increases total QKV parameters

Q15 [MSQ – 2 marks]

After zero-weight initialization (no bias), performing one backpropagation update with SGD, which is true for $W^{(1)}$ (first hidden layer weights)?

- A. It remains zero after the first update
- B. Different neurons receive different gradient updates
- C. All elements are zero after the first update
- D. Entries are either all positive or all negative

Q16 [SA – 2 marks]

A 5×5 grayscale image passes through: Conv(3×3 , 1 filter, stride 1, no padding, no bias) $\rightarrow$ ReLU $\rightarrow$ MaxPool(2×2 , stride 2). Total learnable parameters?

Q17 [SA – 2 marks]

Compute convolution output (stride 1, no padding):

$$X = \begin{bmatrix} 3 & 1 & 2 \\ 0 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}, \quad W = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

Write the full 2×2 output matrix.

Q18 [SA – 2 marks]

CBOV model. Predicted probability for the correct word = 0.25. Compute the cross-entropy loss (natural log). Round to 2 decimal places.

Q19 [SA – 2 marks]

Skip-gram, window size $d = 2$ (each side). Sentence: “the cat sat on mat” (5 words). How many unique (target, context) pairs?

Q20 [SA – 2 marks]

RNN: $s_t = \sigma(Ux_t + Ws_{t-1})$, $U = 1$, $W = 1$, $s_0 = 0$. Input $x_1 = 0$. Find s_1 .

Q21 [SA – 2 marks]

LSTM: $x_t \in \mathbb{R}^4$, $h_t \in \mathbb{R}^5$, $\hat{y}_t \in \mathbb{R}^3$, $T = 8$, with bias. Total trainable parameters?

Q22 [SA – 2 marks]

Transformer: $t = 32$, $d_{\text{model}} = 64$, $h = 2$, $d_v = 16$. Total elements in the attention output for a single head?

Q23 [SA – 2 marks]

FFN: $d_{\text{model}} = 64$, $d_{ff} = 256$, with bias. Total FFN parameters?

Q24 [SA – 1 mark]

A 6×6 grayscale image. Conv with 4 filters of size 3×3 , stride 2, padding 1. What is the output **spatial** size (height or width)?

SOLUTIONS – Practice Paper 1

Q1: B. $P = (F - 1)/2$. From $L = 1 + (L - F + 2P)/1 \Rightarrow 2P = F - 1$.

Q2: D. $f'(x)$ at $x = 1$: A: $1/(x \ln 10) = 0.434$, B: $2/x = 2$, C: $1/(2\sqrt{x}) = 0.5$, D: $3x^2 = 3$. Steepest = x^3 .

Q3: C. All pre-activations = 0 $\rightarrow \sigma(0) = 0.5$ everywhere $\rightarrow$ output = 0.5.

Q4: A. Hidden outputs = 0.5, but output layer weights = 0, so $\text{linear}(0 \times 0.5) = 0$.

Q5: C. S1 is false (LSTM has $\sim 4x$ more params). S2 is true.

Q6: B. Updates per epoch = $10000/50 = 200$. Decay at update 200: $\eta = 0.1 \times 0.5 = 0.05$.

Q7: C. Diagonal = 0 (token can attend to itself). Sum = 0.

Q8: C. Each head has its OWN W_Q, W_K, W_V .

Q9: A. Midpoint at $x = -b/w$. Increasing $b \rightarrow x$ shifts more negative (left).

Q10: A. For (0,0): $h = \text{ReLU}([0, 0]) = [0, 0]$, $\hat{y} = 0 < 0.5 \rightarrow$ class 0. For (1,0): $h = \text{ReLU}([1, -1]) = [1, 0]$, $\hat{y} = 1 > 0.5 \rightarrow$ class 1. For (0,1): $h = \text{ReLU}([-1, 1]) = [0, 1]$, $\hat{y} = 1 > 0.5 \rightarrow$ class 1. For (1,1): $h = \text{ReLU}([0, 0]) = [0, 0]$, $\hat{y} = 0 < 0.5 \rightarrow$ class 0. All correct.

Q11: B, C, E. Overfitting problem. Adam is an optimizer (doesn't fix it). More layers worsens it.

Q12: A, D. NAG uses momentum with look-ahead. Adam's first moment m_t is momentum-like. AdaGrad/RMSProp/AdaDelta are adaptive LR only.

Q13: B, C, E. SVD: reduces sparsity, captures latent semantics, produces dense embeddings.

Q14: A, C. B is false (heads have own projections). D is false (with fixed d_{model} , $d_k = d_{\text{model}}/h$ shrinks as h grows, total QKV stays same).

Q15: D. Symmetry not broken $\rightarrow$ all neurons get same gradient $\rightarrow$ entries all positive or all negative.

Q16: $3 \times 3 = 9$ kernel params. No bias. Pool: 0. Total = **9**.

Q17: Output size = $1 + (3 - 2)/1 = 2$.

(0, 0): $3(1) + 1(-1) + 0(0) + 2(1) = 3 - 1 + 2 = 4$.

(0, 1): $1(1) + 2(-1) + 2(0) + 1(1) = 1 - 2 + 1 = 0$.

(1, 0): $0(1) + 2(-1) + 1(0) + 0(1) = -2$.

(1, 1): $2(1) + 1(-1) + 0(0) + 3(1) = 2 - 1 + 3 = 4$.

$$\begin{bmatrix} 4 & 0 \\ -2 & 4 \end{bmatrix}.$$

Q18: $L = -\ln(0.25) = \ln(4) \approx \mathbf{1.39}$.

Q19: 5 words, $d = 2$: $2 + 3 + 4 + 3 + 2 = \mathbf{14}$.

Q20: $s_1 = \sigma(1 \cdot 0 + 1 \cdot 0) = \sigma(0) = \mathbf{0.5}$.

Q21: $n = 4, d = 5, k = 3$. $4d(n + d + 1) + k(d + 1) = 4(5)(10) + 3(6) = 200 + 18 = \mathbf{218}$. T irrelevant.

Q22: $t \times d_v = 32 \times 16 = \mathbf{512}$.

Q23: $2(64)(256) + 256 + 64 = 32768 + 320 = \mathbf{33088}$.

Q24: $1 + (6 - 3 + 2)/2 = 1 + 5/2 = 1 + 2 = \mathbf{3}$.